SHADOWNET NEXUS v5.0

Unified Live Forensic Intelligence Architecture

ShadowNet Nexus v5.0 is a **live endpoint forensic intelligence platform** designed to detect, correlate, preserve, and report anti-forensic activities in real time.

Unlike traditional forensic tools that require disk imaging and post-incident analysis, ShadowNet:

- Runs on a live system
- Detects attacks while they happen
- Captures evidence before destruction
- Correlates findings using AI
- Generates court-ready reports

2. CORE PROBLEM STATEMENT

Modern attackers use anti-forensics techniques such as:

• Log deletion (wevtutil cl)
• Disk wiping (sdelete, shred)
• Timestamp manipulation (timestomp)
• Hidden encrypted volumes (VeraCrypt)
• Scripted automation (bot keystrokes)

Traditional tools fail because:

- They rely on OS APIs (which attackers manipulate)
- They analyze after the attack
- They do not correlate cross-layer evidence

ShadowNet solves this by combining:

- Raw disk parsing
- Statistical entropy analysis
- Behavioral keystroke profiling
- AI-based correlation
- Cryptographic evidence sealing

3. COMPLETE SYSTEM FLOW (STEP-WISE)

STEP 1 — System Initialization

Main File: shadownet_nexus_v5.py

• Load config.yaml
• Initialize detection modules
• Initialize bounded queues
• Register threads with Watchdog
• Start Health API

System enters continuous monitoring mode.

STEP 2 — Parallel Detection Engines Start

Each engine runs in a separate thread:

• NTFS Artifact Parser
• Entropy Map Scanner
• Realtime Behavior Monitor
• Process Monitor
• Proactive Evidence Collector

Each thread sends heartbeat to Watchdog.

STEP 3 — Threat Detection

When a suspicious event occurs:

Example:
wevtutil cl Security

Process Monitor detects keyword → fires callback.

Simultaneously:

• NTFS Parser checks MFT inconsistencies
• Entropy Scanner checks unallocated space
• Behavior Monitor analyzes typing pattern

STEP 4 — Emergency Snapshot

Before evidence is destroyed:

EmergencySnapshotEngine captures:

• Event logs
• Running processes
• Network connections
• Registry hives
• Memory (if admin)

Snapshot stored in:
evidence/emergency_snapshots/SNAP-XXXX

STEP 5 — Evidence Chain Hardening

EvidenceChain.generate_manifest():

• SHA-256 hash all files
• Create manifest.json
• Request RFC 3161 trusted timestamp
• Lock files (read-only)

Ensures tamper-proof evidence.

STEP 6 — AI Analysis

Gemini Command Analyzer:

• Decodes obfuscated commands
• Classifies intent
• Maps MITRE ATT&CK TTP
• Assigns confidence score

Protected by:

• Rate limiter
• Circuit breaker
• TTL caching

STEP 7 — Narrative Correlation

Forensic Narrative Engine:

Every 60 seconds:

• Drain all finding queues
• Combine NTFS + Entropy + Behavior + Process
• Generate unified incident timeline
• Produce JSON + Markdown report

STEP 8 — Alert & SIEM Forwarding

AlertManager:

• Console
• Slack
• Email
• Discord
• SMS
• Webhook

SIEMIntegration:

• Splunk (HEC)
• Elastic (ECS)
• QRadar
• Syslog

STEP 9 — Continuous Monitoring

Watchdog:

• Monitors thread health
• Restarts crashed modules
• Uses exponential backoff

HealthAPI:

• Exposes system status at localhost:8765

TERMINOLOGY & CONCEPTS

Shannon Entropy

Measure of randomness in data.
Formula:
$H = -\Sigma\ p(x) \log_2 p(x)$

High entropy (~8 bits/byte) → encrypted/random data.

Kolmogorov-Smirnov Test

Statistical test comparing distribution to uniform distribution.
Used to confirm true randomness.

Chi-Squared Test

Tests expected vs observed byte frequency.
Validates cryptographic randomness.

Monte Carlo Pi Test

Estimates π from byte distribution.
Random data converges near 3.14159.

Timestomping

Modifying file timestamps to hide activity.

Compare:
$STANDARD_INFORMATION vs $FILE_NAME

USA Fixup (Update Sequence Array)

NTFS integrity mechanism preventing sector corruption.

Inter-Key Latency (IKI)

Time between two keystrokes.
Humans show variance.
Bots show constant timing.

Gaussian Mixture Model (GMM)

Machine learning model to detect bimodal vs unimodal typing patterns.

RFC 3161 Trusted Timestamp

Cryptographic proof that evidence existed at specific time.

Circuit Breaker

Prevents cascading API failures by disabling repeated failed calls temporarily.

MITRE ATT&CK

Global framework mapping attacker techniques (TTPs).

Example:
T1070.006 → Timestomping

7. ARCHITECTURAL DESIGN PRINCIPLES

• Parallel processing
• Loose coupling (queue-based)
• Fail-safe fallback
• Zero-trust metadata parsing
• Cryptographic immutability
• Multi-layer statistical validation
• AI-assisted correlation
• Enterprise SOC integration
• Self-healing resilience

8. ADVANTAGES OF v5 ARCHITECTURE

• No disk imaging required
• Live anti-forensics detection
• Statistical + behavioral + metadata intelligence
• Court-admissible evidence chain
• Enterprise-ready
• AI-based unified narrative
• Auto-restart resilience
• Multi-channel alerting

9. POTENTIAL VIVA / JURY Q&A

Q1: Why not use standard forensic tools?

Because they:
• Require disk imaging
• Analyze after the attack
• Depend on OS APIs

ShadowNet reads raw disk structures and detects in real time.

Q2: How do you detect hidden encrypted volumes?

Using:

• Shannon entropy
• K-S test
• Chi-Squared
• Monte Carlo Pi

Only when all pass → confirmed hidden volume.

Q3: How do you detect timestomping?

Compare:

$SI.modified vs $FN.modified

If SI < FN → backdating confirmed.

Q4: What ensures evidence integrity?

• SHA-256 hashing
• Manifest generation
• RFC 3161 trusted timestamp
• Read-only locking

Q5: What happens if Gemini API fails?

Circuit breaker activates.
System switches to rule-based fallback.
No detection stops.

Q6: How do you prevent alert storms?

AlertDeduplication module suppresses identical alerts within time window.

Q7: How is system resilient?

Watchdog monitors threads.
If crash → restart with exponential backoff.

Q8: What makes this different from EDR?

EDR:
• Signature-based
• Process-focused

ShadowNet:
• Raw NTFS parsing
• Statistical disk intelligence
• Behavioral biometrics
• AI narrative synthesis
• Cryptographic evidence chain

Q9: Can it run on production servers?

Yes.
Admin required for NTFS & entropy modules.
Non-admin mode still supports process & behavior monitoring.

Q10: What is overall innovation?

Simultaneous:

• Raw filesystem intelligence
• Statistical disk analysis
• Behavioral profiling
• AI synthesis
• Cryptographic chain-of-custody

All live. No imaging.

D. Logical Architecture Diagram (Textual)

Code

```
                ┌──────────────────┐
                │   shadownet_nexus_v5   │
                └──────────────────┘
                          │
                          │
     ┌────────────┬────────┼────────────┬────────────┐
     │            │        │            │            │
 NTFS Parser   Entropy Scan  Behavior   Process Monitor  Evidence Collector
     │            │        │            │            │
     └────────────┴────────┼────────────┴────────────┘
                          │
                          │
                    Narrative Queues
                          │
                          ▼
                Forensic Narrative Engine
                          │
                          ▼
                  Gemini AI Synthesis
                          │
        ┌────────────┬────────────┬────────────┐
        │            │            │            │
   Incident Report  SIEM Forward  Alert System   Evidence Vault
```

Code

```
            SYSTEM START
                  |
                  ▼
        Load Config & Initialize Threads
                  |
                  ▼
        ┌──────────────────────┐
        │  Detection Layer (Eyes)  │
        └──────────────────────┘
                  |
   ┌──────────────┼──────────────┐
   ▼              ▼              ▼
Process        NTFS Parser     Entropy Scan
Monitor                           |
   |              |              |
   └──────────────┼──────────────┘
          ▼              ▼
      Evidence Collector   Behavior Monitor
          |
          ▼
   Emergency Snapshot Engine
          |
          ▼
     Evidence Chain (SHA-256 + TSA)
          |
          ▼
      AI Command Analyzer
          |
          ▼
  Forensic Narrative Engine
          |
          ▼
   Incident Report Generator
          |
   ┌──────┼──────┐
   ▼              ▼
Alert Manager     SIEM Integration
          |
          ▼
   Continuous Monitoring (Watchdog)
```